Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity. Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an ”as is” basis, without warranty. Any error report is appreciated.

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WEEK III

5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

- **Wave function and probability / Ex. 5.48**
The wave function of a particle in a one-dimensional box of width L is

$$\psi(x) = A\sin(\pi x/L)$$

If we know the particle must be somewhere in the box, what must be the value of A?

Solution
The condition means that the probability associated with the wavefunction of finding the particle in the box is unitary, i.e., the wave function is normalized

$$\int_0^L \mathrm{d}x\, |\psi(x)|^2 = A^2 \int_0^L \mathrm{d}x\, \sin^2(\pi x/L) = 1.$$

Strategy: (i) variable transformation $\pi x/L \to u$, (ii) use the trigonometric identities $\sin^2(x) + \cos^2(x) = 1$ and that $\cos(2x) = \cos^2(x) - \sin^2(x)$, thus yielding $2\sin^2(x) = 1 - \cos(2x)$.

One obtains

$$A^2 \int_0^L \mathrm{d}x\, \sin^2(\pi x/L) = A^2 \frac{L}{\pi} \int_0^\pi \mathrm{d}u\, \sin^2(u) = \frac{A^2}{2}\frac{L}{\pi} \int_0^\pi \mathrm{d}u\, \Big(1 - \cos(2u)\Big) = 1$$

the first term integrates to π while the second integrates to zero by symmetry (integral of a periodic function over a period, if not obvious, substitute $2u \to w$, resulting in the integral of $\cos(w)$ between 0 and 2π...) yielding

$$A^2 \frac{L}{2} = 1 \qquad \text{or} \qquad A = \sqrt{\frac{2}{L}}.$$

Note that the overall phase of the wavefunction is arbitrary, so any choice of the form $A' = Ae^{i\phi}$ with a real ϕ is equally valid, but it's customary to choose the wavefunction to be real if possible.

6. QUANTUM MECHANICS II

- **Wave function normalization / Ex. 6.4**
Normalize the wave function $\psi(x,t) = Ae^{i(kx-\omega t)}$ in the region $x = [0, a]$.

Solution
The normalization condition is

$$1 = \int_0^a \mathrm{d}x\, |\psi(x,t)|^2 = A^2 \int_0^a \mathrm{d}x = A^2 a,$$

yielding $A = \sqrt{1/a}$.

- **Wave function normalization... actually notable integrals! / Ex. 6.5**
Normalize the wave function $\psi(r) = Are^{-r/a}$ in the region $r = [0, \infty]$ where a and A are constants.
Integrals: $I_n = \int x^n e^{-x/\alpha}$; $I_0 = -\alpha e^{-x/\alpha}$; $I_1 = -(\alpha^2 + \alpha x)e^{-x/\alpha}$; $I_2 = -(2\alpha^3 + 2\alpha^2 x + \alpha x^2)e^{-x/\alpha}$; and in general, all other integrals can be obtained recursively as $I_{n+1} = \alpha^2 \frac{d}{d\alpha} I_n$.

Solution
The normalization condition is found using integral I_2 with $\alpha \to a/2$

$$1 = \int_0^\infty \mathrm{d}r\, |\psi(r)|^2 = A^2 \int_0^\infty \mathrm{d}r\, r^2 e^{-2r/a} = A^2 \Big[-\left(\frac{1}{4}a^3 + \frac{1}{2}a^2 r + \frac{1}{2}ar^2\right) e^{-r/a} \Big]_0^\infty = A^2 \frac{1}{4}a^3,$$

yielding $A = 2a^{-3/2}$.

- **Probability of finding the particle in a region / Ex. 6.11**
  A wave function has the value $\psi(x) = A\sin(x)$ in $x = [0, \pi]$ but zero elsewhere. Normalize the wave function and find the probability that the particle is (a) between $x = 0$ and $x = \pi/4$ and (b) between $x = 0$ and $x = \pi/2$.

  Solution
  First of all, let us find A by normalizing the wave function

$$1 = \int_{-\infty}^{\infty} |\psi(x)|^2 = A^2 \int_0^{\pi} \sin^2(x) = A^2 \frac{1}{2} \int_0^{\pi} \mathrm{d}x \Big[1 - \cos(2x)\Big] = \frac{\pi}{2}A^2 - \frac{1}{4}A^2 \Big[\sin(2x)\Big]_0^{\pi} = \frac{\pi}{2}A^2$$

  yielding

$$A = \sqrt{\frac{2}{\pi}}.$$

  The probability to find the particle in an interval $[x_{\mathrm{min}}, x_{\mathrm{max}}]$ is given by

$$P = \int_{x_{\mathrm{min}}}^{x_{\mathrm{max}}} \mathrm{d}x |\psi(x)|^2.$$

  The integral then yields

$$P = \frac{1}{\pi} \int \mathrm{d}x \Big(1 - \cos(2x)\Big) = \frac{1}{\pi}\Big[x - \frac{1}{2}\sin(2x)\Big]_{x_{\mathrm{min}}}^{x_{\mathrm{max}}}. \tag{1}$$

  Hence for (a) one gets

$$P = \frac{1}{4} - \frac{1}{2\pi} = 0.0908,$$

  whereas for (b) one gets

$$P = \frac{1}{2}.$$

- **Classical and quantum probability in the infinite square-well / Ex. 6.13**
  Determine the average value of $|\psi_n(x)|^2$ inside the well for the infinite square-well potential for $n = \{1, 5, 10, 100\}$. Compare these averages with the classical probability of detecting the particle inside the box.

  Solution
  The wave function for a particle in the infinite square-well potential of width L is given by

$$\psi(x) = \sqrt{\frac{2}{L}}\sin(k_n x) \qquad k_n = n\frac{\pi}{L}.$$

  Recall that this is derived by

  (i) imposing boundary conditions $\psi(0) = \psi(L) = 0$, which determines the phase and allowed wavenumbers k_n
  (ii) finding the proper normalization.

  The average value of the wave function is given by

$$\big\langle |\psi_n(x)|^2 \big\rangle = \frac{\int_0^L \mathrm{d}x |\psi_n(x)|^2}{\int_0^L \mathrm{d}x} = \frac{1}{L}\int_0^L \mathrm{d}x |\psi_n(x)|^2 = \frac{2}{L^2}\int_0^L \mathrm{d}x \sin^2(k_n x) = \frac{2}{L^2}\left[\frac{x}{2} - \frac{1}{4n}\sin\left(\frac{2nx}{L}\right)\right]_0^L = \frac{1}{L},$$

  which is *independent* on n and is the same as the classical probability, which is uniform throughout the well. Note that the numerator corresponds to the normalization of the wave function, so one could have obtained trivially that $1/L$ is the correct result.

  Instead, the quantum mechanical probability is given by

$$|\psi_n(x)|^2 = \frac{2}{L}\sin^2(k_n x)$$

  which is not uniform.

- **Infinite square-well potential approximation for a nanowire / Ex. 6.15**
  We can approximate an electron moving in a nanowire (a small, thin wire) as a one-dimensional infinite square-well potential. Let the wire be $L = 2.0\,\mu\text{m}$ long. The nanowire is cooled to a temperature $T = 13\,\text{K}$, and we assume the electron's average kinetic energy is that of gas molecules at this temperature $\langle K \rangle = \frac{3}{2} k_B T$. (a) What are the three lowest possible energy levels of the electrons? (b) What is the approximate quantum number of electrons moving in the wire?

  Solution
  The energy levels of an infinite square-well potential are given by

  $$E_n = n^2 \frac{h^2}{8 m_e L^2} = n^2 \frac{(hc)^2}{8(m_e c^2) L^2} = 9.40 \times 10^{-8}\,\text{eV} \cdot n^2,$$

  where $hc = 1240\,\text{eV nm}$ and $m_e c^2 = 511\,\text{keV}$ was used. Hence, the first three levels are given by

  - $E_1 = 9.40 \times 10^{-8}\,\text{eV}$,
  - $E_2 = 3.76 \times 10^{-7}\,\text{eV}$,
  - $E_3 = 8.46 \times 10^{-7}\,\text{eV}$.

  At $T = 13\,\text{K}$ the electron's average kinetic energy is given by

  $$\langle K \rangle = \frac{3}{2} k_B T = 1.68 \times 10^{-3}\,\text{eV},$$

  where the Boltzmann constant $k_B = 1.381 \times 10^{-23}\,\text{J/K}$. with the conversion $1\,\text{J} = 1.602 \times 10^{-19}\,\text{eV}$ or directly $k_B = 8.617 \times 10^{-5}\,\text{eV/K}$.

  Substituting this value for E_n and solving for n yields $n = 133.68 \approx 134$.

- **Emission spectra for an electron in an infinite square-well potential / Ex. 6.21**
  An electron is trapped in an infinite square-well potential of width $L = 0.70\,\text{nm}$. If the electron is initially in the $n = 4$ state, what are the various photon energies that can be emitted as the electron jumps to the ground state $n = 1$?

  Solution
  he energy levels of an infinite square-well potential are given by

  $$E_n = n^2 \frac{h^2}{8 m_e L^2} = n^2 \frac{(hc)^2}{8(m_e c^2) L^2},$$

  so that the ground state energy is given by

  $$E_1 = \frac{h^2}{8 m_e L^2} = 0.7676\,\text{eV},$$

  where $hc = 1240\,\text{eV nm}$ and $m_e c^2 = 511\,\text{keV}$ was used.

  The photon energies for all possible transitions (disregarding any transition rules) are

  - $E_4 - E_3 = 5.37\,\text{eV}$
  - $E_4 - E_2 = 9.21\,\text{eV}$
  - $E_4 - E_1 = 11.5\,\text{eV}$
  - $E_3 - E_2 = 3.84\,\text{eV}$
  - $E_3 - E_1 = 6.14\,\text{eV}$
  - $E_2 - E_1 = 2.30\,\text{eV}$

- **Biological molecules as particles in a box**
  Many biological molecules contain long chains of carbon atoms, which can be approximated as a one-dimensional potential well for the π electrons (the second electron in every double bond) that are delocalized along the chain. Consider β-carotene, a molecule that gives the orange color of carrots for example. The reason for the orange color is that this molecule absorbs light in the blue-green range (around 450 nm) and reflects only wavelengths that we see as orange-red. Let's estimate the absorption spectrum of β-carotene based on the particle in the box model, using that the carbon chain has a length of approximately $L = 2.4\,\mathrm{nm}$, and there are 22 delocalized electrons in the molecule.

  Solution
  The energy levels of the box are given by

$$E_n = \frac{h^2}{8m_\mathrm{e}L^2}\, n^2.$$

In the ground state, the 22 electrons fill the 11 lowest energy levels (because of the Pauli principle there can only be one electron in each state, and every level of the box has two states with different spin). The lowest energy excitation occurs when an electron on the $n_i = 11$ level absorbs a photon, and is excited to the $n_f = 12$ level. The energy difference corresponding to this excitation is

$$\Delta E = \frac{h^2}{8m_\mathrm{e}L^2}(n_f^2 - n_i^2) = 2.41 \times 10^{-19}\,\mathrm{J},$$

where $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$ and $m_\mathrm{e} = 9.109 \times 10^{-31}\,\mathrm{kg}$ is the electron mass. This energy difference corresponds to a photon wavelength of

$$\lambda = \frac{hc}{\Delta E} = 826\,\mathrm{nm}.$$

This is in the infrared range. We can also calculate the wavelength corresponding to the second lowest energy excitation with $n_i = 10$ and $n_f = 12$, which gives $\lambda = 432\,\mathrm{nm}$, much closer to the measured value. A comparison of all the spectral lines shows that this approximation is not very accurate, and getting such a close result is merely a coincidence. Nevertheless, this simplified picture helps to understand trends, such as longer molecules absorb light at longer wavelengths.

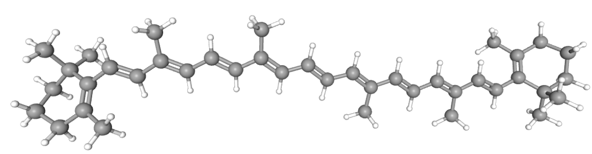

Figure 1: National Center for Biotechnology Information. PubChem Compound Summary for CID 5280489, Beta-Carotene. `https://pubchem.ncbi.nlm.nih.gov/compound/Beta-Carotene`. Accessed Mar. 10, 2025.

- **Particle in a 3D box**

A particle is confined to an infinite potential well in 3 dimensions with a cubic shape of size $L \times L \times L$. Its ground state wavefunction has the form

$$\psi(x,y,z) = A \sin\left(\frac{\pi}{L}x\right) \sin\left(\frac{\pi}{L}y\right) \sin\left(\frac{\pi}{L}z\right). \tag{2}$$

(a) Calculate the value of A that normalizes the wavefunction. (b) Calculate the probability of finding the particle in the corner of the box where $x, y, z < L/3$.

Solution

(a) The normalization condition now involves integrating over all of the spatial coordinates:

$$1 = \iiint_{[0,L]^3} dx\, dy\, dz\, |\psi(x,y,z)|^2 = A^2 \iiint_{[0,L]^3} dx\, dy\, dz\, \sin^2\left(\frac{\pi}{L}x\right) \sin^2\left(\frac{\pi}{L}y\right) \sin^2\left(\frac{\pi}{L}z\right).$$

The wavefunction is a product of three terms, each of which only depends on one of the coordinates, so the integral can be separated into a product of integrals.

$$1 = A^2 \left(\int_0^L dx \sin^2\left(\frac{\pi}{L}x\right)\right) \left(\int_0^L dy \sin^2\left(\frac{\pi}{L}y\right)\right) \left(\int_0^L dz \sin^2\left(\frac{\pi}{L}z\right)\right) = A^2 \left(\int_0^L dx \sin^2\left(\frac{\pi}{L}x\right)\right)^3 = A^2 \left(\frac{L}{2}\right)^3,$$

from which

$$A = \left(\frac{2}{L}\right)^{3/2}.$$

(b) To find the probability, we integrate the absulute square of the wavefunction for the region of interest:

$$P = \iiint_{[0,L/3]^3} dx\, dy\, dz\, |\psi(x,y,z)|^2 = \left(\frac{2}{L}\right)^3 \left(\int_0^{L/3} dx \sin^2\left(\frac{\pi}{L}x\right)\right)^3 = 8\left(\frac{1}{6} - \frac{\sqrt{3}}{8\pi}\right)^3 \approx 0.00747.$$